



US012384900B2

(12) **United States Patent**
Kamphus et al.

(10) **Patent No.:** **US 12,384,900 B2**
(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **RECYCLING OF SUPERABSORBENT FIBERS VIA UV IRRADIATION IN FLOW SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 735 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/591,648**

(22) Filed: **Feb. 3, 2022**

(65) **Prior Publication Data**
US 2022/0267558 A1 Aug. 25, 2022

Related U.S. Application Data

(60) Provisional application No. 63/151,862, filed on Feb.
22, 2021.

(51) **Int. Cl.**
C08J 3/28 (2006.01)
C08J 11/10 (2006.01)

(52) **U.S. Cl.**
CPC **C08J 11/10** (2013.01); **C08J 3/28**
(2013.01); **C08J 2333/02** (2013.01)

(58) **Field of Classification Search**
CPC C08J 11/10; C08J 3/28; C08J 2333/02;
C08J 11/04
USPC 521/40.5
See application file for complete search history.

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(57) **ABSTRACT**

Superabsorbent fibers (SAF) in a feed stream is converted
with UV irradiation into soluble polymers in a flow system.
The UV total energy used to convert SAF into soluble
polymers is less than about 50 MJ/kg SAF.

25 Claims, No Drawings

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1

RECYCLING OF SUPERABSORBENT FIBERS VIA UV IRRADIATION IN FLOW SYSTEM

FIELD

The present disclosure generally relates to recycling of superabsorbent polymer (SAF) using UV irradiation in a flow system and at short residence times. More specifically, a feed stream comprising SAF is fed into the flow system and a product stream is produced, which comprises essentially soluble polymers. The concentration of SAF in the feed stream is greater than about 1 wt %, and the UV total energy used to convert SAF to soluble polymers is less than about 50 MJ/kg SAF.

BACKGROUND

Recycling of absorbent hygiene products (AHPs) (i.e., baby diapers, feminine protection pads, and adult incontinence pads) is good for the environment and needed to achieve the sustainability goals of many consumer companies. These goals are about using 100% recycled materials and having zero consumer and manufacturing waste go to landfill. In addition to these goals, successful recycling benefits the environment, stimulates the economy, improves people's health and water quality, and generates energy needed by consumers in developing regions of the world.

The major component in AHPs is typically the superabsorbent polymer (SAP), whereas other components are adhesives, cellulose fibers, polyethylene, polypropylene, and polyester. SAP is a water-absorbing, water-swellaable, and water-insoluble powdered solid which is a crosslinked and partially neutralized homopolymer of glacial acrylic acid. SAP has an exceptionally high ability to absorb aqueous liquids, such as contaminated water or urine. An alternative to poly(acrylic acid) based SAP are superabsorbent fibers (SAF). These fibers generally comprise a relatively large amount of comonomers (when compared with superabsorbent polymer particles). Therefore, similar to SAP, there is also a need to recycle SAF used in AHPs.

Recycling of AHPs involves cleaning of the AHPs from the soils accumulated during their use and separating the various components into recycled material streams. More specifically, the recycled SAF material stream can be used in applications less demanding than AHPs (since the recycled SAF has inferior properties compared to virgin SAF; for example, agricultural or horticultural water-retaining agents, and industrial waterproofing agents) and/or can be converted to essentially non-crosslinked, and slightly branched or linear soluble polymers. For example, the soluble polymers can be: 1) used as-is in applications such as water treatment or corrosion inhibition; or 2) esterified and then used in adhesives, coatings, etc.; or 3) used as-is in SAF making such as SAF fiber spinning or 4) transformed into SAF (see item 3)) and blended with virgin SAF. The first two sets of applications are part of the effort to recycle SAF into other products by replacing virgin compounds with compounds derived from recycled SAF, whereas the last two sets of applications are part of the circular economy of SAF, i.e., recycling SAF back to SAF. In all cases, the objective is to achieve the same properties as virgin materials.

While no references have been identified to address degradation of SAF not purely based on acrylic acid, non-limiting examples of processes that produce purified and separated material streams of used SAP from recycled AHPs are disclosed and claimed in U.S. Pat. No. 9,095,853 B2,

2

issued on Aug. 4, 2015; and 9,156,034 B2, issued on Oct. 13, 2015; both assigned to Fater S.p.A, based in Pescara, Italy.

Most SAPs are based on poly(acrylic acid) and are crosslinked network materials. Non-limiting examples of procedures used to produce SAPs from glacial acrylic acid and crosslinkers are disclosed in U.S. Pat. No. 8,383,746 B2, issued on Feb. 26, 2013, and assigned to Nippon Shokubai Co., Ltd, based in Osaka, Japan; and U.S. Pat. No. 9,822,203 B2, issued on Nov. 21, 2017, and assigned to BASF SE, based in Ludwigshafen, Germany.

There are many references on attempts to degrade or de-polymerize linear polymers and only a few references on efforts to depolymerize SAPs. The typical forms of energy used in these efforts (either as single forms of energy or in combination) are ultrasonic, UV, mechanical (i.e., in the presence of extensional/elongational forces; examples are: Caruso, M. M., et al., *Chem. Rev.*, 109 (2009), 5755-5798), thermal (example: McNeill, I. C., and Sadeghi, S. M., *Polymer Degrad. Stability*, 29 (1990), 233-246), and microwave. UV is exclusively used with a free-radical generating agent, such as hydrogen peroxide (H_2O_2). Examples of using UV and hydrogen peroxide are described in Kaczmarek, H., et al., *Die Angew. Makrom. Chemie*, 261/262 (1998), 109-121, and Mierzwa, J. C., and Rodrigues, R., *Adv. Oxidation Proc. Wastewater Treatm.*, Chapter 2 (2018), 13-48. An example of using ultrasound, UV, and H_2O_2 , is described in Prajapat, A. L., and Gogate, P. R., *Ultrasonics Sonochemistry*, 32 (2016), 290-299. The main themes from all these references on the degradation or de-polymerization of linear polymers are that the preferential scission is at the mid-point of the polymer chain, the higher molecular weight chains are degraded at a higher rate than the lower molecular weight chains, and there is a minimum molecular weight below which degradation or de-polymerization does not occur.

More recently, Li, X., and Cui, Y., *J. Appl. Polym. Sci.*, 108 (2008), 3435-3441; and Shukla, N. B., and Madras, G., *J. Appl. Polym. Sci.*, 125 (2011), 630-639, have attempted to degrade poly(acrylic acid)-based SAP using UV irradiation in static systems. In the first work, the crosslinker was N,N-Methylenebisacrylamide (NMBA) and in the second work it was ethylene glycol dimethacrylate (EGDMA). Weight loss, swelling capacity, and residual weight fraction are the three parameters used to follow the degradation progress over the irradiation time. The time scale of UV irradiation required to achieve zero swelling capacity, or zero residual weight fraction was reported to be about 120 min, which is not commercially viable. At a fixed temperature, close to room temperature (e.g., 30° C.), UV irradiation power higher than 1500 W does not increase the weight loss any further, whereas, at a higher temperature (e.g., 90° C.), UV irradiation power higher than 1500 W increases the weight loss slowly.

Accordingly, there is a need to recycle AHPs and their major component, which is SAF for some AHPs. For the recycling of SAF, there is a need to degrade recycled SAF into soluble polymers, in short time scale; with low energy and power per unit mass of SAF; and at mild conditions, such as room temperature, thus avoiding chemical decomposition such as decarboxylation of the degraded SAF. The requirement for low energy per unit mass of SAF stems from the fact that the recycling of SAF and its degradation to soluble polymers is beneficial only if the energy spent during the converting of SAF to soluble polymers is less than, e.g., that used to make fossil-derived soluble polymers of same kind. For acrylic acid (petro-AA) from propylene, the required energy is about 50 MJ/kg AA. The soluble poly-

mers produced from recycled SAF can then be incorporated back into virgin SAF (thus increasing its recycled content and supporting the circular economy of SAF) and/or derivatized into materials for other applications, such as, adhesives, coatings, water treatment, fabric care, etc.

SUMMARY

In one embodiment of the present disclosure, a method for degrading superabsorbent fibers (SAF) to soluble polymers is presented, wherein the soluble polymers comprise from 5 weight-% to 75 weight-% of polymerized acrylic acid monomer units preferably from 10 weight-% to 75 weight-%, or from 10 weight-% to 70 weight-%, based on the total weight of the soluble polymers. As used herein, the term "acrylic acid monomer units" excludes derivatives of acrylic acid monomers, such as esters and amides, and also excludes methacrylic acid monomer units and salts thereof.

The method comprises flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV, and producing a product stream comprising the soluble polymers at the end of the irradiation zone; wherein the feed stream comprises the SAF at a concentration greater than about 1 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; and wherein the UV used to convert SAF to soluble polymers requires a UV total energy of less than about 50 MJ/kg SAF.

Also, a method for degrading superabsorbent fibers (SAF) to soluble polymers is presented, wherein the soluble polymers comprise from 5 weight-% to 75 weight-%, of polymerized acrylic acid monomer units preferably from 10 weight-% to 75 weight-%, or from 10 weight-% to 70 weight-% based on the total weight of the soluble polymers. The method comprises flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV, and producing a product stream comprising soluble polymers at the end of the UV irradiation zone; wherein the feed stream comprises water and SAF at a concentration greater than about 1 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; wherein the UV used to convert SAF to soluble polymers requires a UV total energy of less than about 16 MJ/kg SAF; and wherein the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol.

The disclosure further relates to a method for degrading superabsorbent fibers (SAF) to soluble polymers, wherein the soluble polymers comprise from 5 weight-% to 75 weight-% of polymerized acrylic acid monomer units, preferably from 10 weight-% to 75 weight-%, or from 10 weight-% to 70 weight-% based on the total weight of the soluble polymers. The method comprises flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV, and producing a product stream comprising soluble polymers at the end of the UV irradiation zone; wherein the feed stream comprises water and SAF at a concentration greater than about 5 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; wherein the UV used to convert SAF to soluble polymers requires a UV total energy of less than about 16 MJ/kg SAF; and wherein the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol.

The soluble polymers may be water soluble. The soluble polymers preferably have a solubility in water at 25° C. of more than 5 g of soluble polymer per 100 g water, or more

than 15 g of soluble polymer per 100 g water, or from 20 g to 120 g of soluble polymer per 100 g water or from 35 g to 100 g of soluble polymer per 100 g water.

The soluble polymers obtained by the method of the present disclosure can be subjected to the NMR Content Method set out below. The NMR Content Method can be used to determine the ratio on a molar basis of protons of functional groups with different NMR signals such as alkene terminal moieties, alkoxy groups ($-\text{O}-\text{CHR}1-$ with R1 one of the group H, alkyl, aryl, heteroaryl, alkoxy or halogene), aliphatic groups ($-\text{CHR}1-$ with R1 one of the group H, alkyl, aryl, heteroaryl or halogene) and/or other ^1H -NMR active groups. The NMR Content Method also enables the determination of many monomer units other than acrylic acid monomer units, which are comprised in the soluble polymers, given that the spectra obtained by the NMR Content Method allow the determination of various functional groups due to their different NMR signals.

The soluble polymers obtained by the method of the present disclosure may have a ratio 3.6:CH of at least 0.04, as determined by the NMR Content Method described herein. The soluble polymers may have a ratio 3.6:CH of at least 0.1, or at least 0.2 as determined by the NMR Content Method described herein. The soluble polymers may have a ratio 3.6:CH of not more than 0.7 as determined by the NMR Content Method described herein.

The soluble polymers obtained by the present disclosure may have a content "%3.6 ppm" of at least 5%, as determined by the NMR Content Method described herein. The soluble polymers may have a content "%3.6 ppm" of at least 10%, or at least 15%, as determined by the NMR Content Method described herein. The soluble polymers may have a content "%3.6 ppm" of not more than 75%, as determined by the NMR Content Method described herein.

The soluble polymers obtained by the method of the present disclosure may have a content of alkene, "% alkene", of not more than 0.31%, as determined by the NMR Content Method described herein.

DETAILED DESCRIPTION

I. Definitions

As used herein, the term "SAF" refers to superabsorbent fibers. The SAF of the present disclosure are capable of absorbing a 0.9 wt % saline solution at 25° C. at least 7 times their dry weight as measured according to the Centrifuge Retention Capacity (CRC) test method described herein, preferably at least 10 times their dry weight. The typical absorption mechanism is osmotic pressure. SAF that absorbs water or aqueous solutions becomes softer and gel-like.

"Superabsorbent fiber" ("SAF") is used herein to refer to superabsorbent polymer material that is in a fibrous form. The superabsorbent fibers have a length and a cross-section. The length is the largest dimension of the fiber when the fiber is or would be laid flat and straight on a surface, such that curves or crimps in the fiber disappear and the fiber becomes an approximately rod-like form. The cross-section is orthogonal to the length. For purposes herein, a fiber is a material that has a largest dimension and smallest dimension, wherein the ratio of largest dimension to smallest dimension is at least 10:1, preferably at least 15:1, even more preferably at least 20:1, i.e. the largest dimension of the superabsorbent fiber (also called the length) is at least 10 times, or at least 15 times, or at least 20 times the smallest dimension of the fiber (also called width). If a superabsorbent fiber has a cross-section that varies along the length of

the fiber, the largest dimension of the cross-section (determined along the length of the fiber) is taken as the width of the fiber when calculating the ratio of largest dimension to smallest dimension.

As used herein, the term “soluble polymers” refers to a non-crosslinked, and either slightly branched or linear polymer, the polymer comprising from 5 weight-% to 75 weight-% of acrylic acid as the monomeric unit and degree of polymerization that can be 2 or higher. For the purposes of the present disclosure, for soluble polymers there will be no difference between a polymer and an oligomer.

As used herein, the term “degradation” refers to the conversion of SAF into soluble polymers via the actions of partial de-polymerization, de-crosslinking, molecular backbone breaking, or any combination of the above actions. For the purposes of the present disclosure, the terms degradation, recycling, and conversion are used interchangeably, as long as they refer to the transformation of SAF to soluble polymers. Also, the degradation essentially preserves the carboxylic groups of the SAF and thus the product soluble polymers contains those carboxylic groups. Note that full de-polymerization of SAF should lead to the monomers which were comprised by the SAF.

As used herein, the term “virgin SAF” refers to SAF produced from virgin monomers, which is the feedstock used today to make SAF. Virgin monomers can be produced from either fossil-derived materials or bio-derived materials (non-limiting examples of bio-materials are: lactic acid, 3-hydroxypropionic acid, glycerin, bio-propylene, carbon dioxide, and sugar). Virgin SAF does not include any recycled SAF above about 1 wt %.

As used herein, the term “used SAF” refers to SAF which has already been produced industrially and/or used commercially, for example, in a baby diaper, feminine pad, adult incontinence pad, or other articles and/or uses. Used SAF can be post-consumer SAF, post-industrial SAF, or combinations of both. Unless otherwise noted, SAF refers to either “used SAF” or “virgin SAF”. As used herein, the term “degraded SAF” refers to SAF which has been degraded to soluble polymers. For the purposes of the present disclosure, the terms “degraded SAF” and “soluble polymers” are used interchangeably.

As used herein, the term “recycled SAF” refers to SAF which contains at least 1 wt % degraded SAF (or equivalently, soluble polymers) that has been incorporated into the SAF while the SAF is being produced from acrylic acid and co-monomers using the typical production method. Thus, the recycled SAF is a blend of virgin SAF and at least 1 wt % degraded SAF.

As used herein, the term “feed stream” refers to a body of fluid that flows in a specific direction and feeds into a flow system.

As used herein, the term “product stream” refers to a body of fluid that is produced from the flow system when the feed stream is fed into it.

As used herein, the term “UV irradiation zone” refers to an area of the flow system that is irradiated by a UV source, for example a Hg lamp. For the purposes of the present disclosure, UV refers to radiation between 10 nm and 400 nm, that includes the vacuum UV (VUV; 10 nm-200 nm), far UV (FUV; UV C; 200 nm-280 nm), middle UV (MUV; UV B; 280 nm-315 nm), and near UV (NUV; UV A; 315 nm-400 nm) ranges.

As used herein, the term “wall shear stress” refers to the shear stress on the wall of the flow system using the viscosity of the feed stream μ , assuming that there is no slip

at the wall. In case the flow system is a circular tube with uniform radius R, the wall shear stress (τ_w) is calculated using the formula:

$$\tau_w = \frac{4\mu\dot{Q}}{\pi R^3},$$

where $\dot{Q}$ is the volumetric flowrate of the feed stream.

As used herein, the terms “viscosity ratio” or “viscosity reduction ratio” refer to the ratio of the viscosity of the product stream to that of the feed stream. The viscosity of the feed stream is typically measured with a parallel plate fixture in oscillatory mode, and the complex viscosity reported typically corresponds to a frequency of 1 rad/s. The real and imaginary part of the complex viscosity represent the dynamic viscosity and the storage viscosity, respectively. For calculating the viscosity ratio, we use the dynamic viscosity i.e. the real part of the complex viscosity. The viscosity of the product stream is measured with either a cup and bob fixture in steady mode or parallel plate fixture in oscillatory mode. When the viscosity is measured with a cup and bob fixture in steady mode the viscosity reported typically corresponds to a shear rate of 4 s^{-1} . These viscosity measurement techniques are well known to those skilled in the art. For the purposes of the present disclosure, the negative of the decadic logarithm of the viscosity ratio indicates the extent of the SAF degradation to soluble polymers in orders of magnitude, as it is accepted by those skilled in the art that the lower the viscosity of a given soluble polymer solution the lower the molecular weight of the soluble polymers is, at a fixed concentration.

As used herein, M_n is the number average molecular weight, in g/mol or equivalently Da, M_w is the weight average molecular weight, in g/mol or equivalently Da, M_z is the z-average molecular weight, in g/mol or equivalently Da, and PDI is the polydispersity index defined as M_w/M_n .

As used herein, the term “ply glue” refers to an adhesive that is used to glue 2 or more plies together. For example, a ply glue for paper towel products is used to glue 2 paper plies together thus forming a 2-ply paper towel product.

II. Feed Stream

Unexpectedly, it has been found that the UV-based degradation of SAF to soluble polymers (i.e., essentially, without decarboxylation) is much more pronounced when the SAF feed stream flows while being irradiated, compared to when the SAF is static while being irradiated, at the same residence time. Without wishing to be bound by any theory, applicants believe that the stresses generated in the flowing feed stream cause stretching of the bonds and either: 1) lower the UV energy required to degrade SAF to soluble polymers; and/or 2) accelerate the kinetics of the SAF degradation.

The typical properties of SAF are mechanical properties, swelling capacity, and centrifuge retention capacity (CRC) as measured according to the test method as described herein. Also, the SAF includes from more than 25 weight-% to less than 95 weight-% of other co-monomers (i.e. other than acrylic acid monomers). Suitable co-monomers are monomeric groups e.g. comprised of/consisting of methyl acrylate, methyl methacrylate, ethyl (meth) acrylate 2-ethyl hexyl (meth) acrylate, or polyacrylamides, polyvinyl alcohol, ethylene maleic anhydride copolymers, polyvinyl ethers, hydroxypropylcellulose, polyvinylmorpholinone, and

polymers and copolymers of vinyl sulfonic acid, polyvinyl amines, polyallyl amines, polyvinylpyrrolidone, and the like. Other suitable polymers in SAF include hydrolyzed acrylonitrile grafted starch, acrylic acid grafted starch, and isobutylene maleic anhydride copolymers and mixtures thereof. The SAF are crosslinked to render the material substantially water insoluble. Preferred monomers are methyl (meth) acrylate, and/or monomers comprising a hydroxylic group, e.g. hydroxyalkyl (meth) acrylates, e.g. hydroxyethyl methacrylate, tripropyleneglycol mono acrylate, 5 glyceryl monoacrylate etc. The SAF can comprise/consist of polymers formed of two kinds of monomers (including acrylic acid) or more preferably of three or more kinds of monomers.

The SAF may have a sodium level as amount of Na in wt % greater than about 8 wt %, or a sodium level as amount of Na in wt % of from 10 wt % to 20 wt %, or from 15 wt % to 18 wt %. In yet another embodiment, the SAF has or a sodium level as amount of Na in wt % below 12 wt % (for the avoidance of doubt, the terms "weight-%", "wt-%" and "wt %" are used herein interchangeably).

The SAF provided in the method may be i) in the form of loose fibers or may be ii) in the form of a nonwoven web, or may be a combination of i) and ii). If the SAF are in the form of a nonwoven web, the nonwoven web may consist of SAF or may comprise SAF. A nonwoven web comprising SAF and provided in the method may comprise at least 50 weight-% of SAF based on the total weight of the nonwoven web, or may comprise at least 60 weight-%, or at least 70 weight-%, or at least 80 weight-%, or at least 90 weight-% of SAF based on the total weight of the nonwoven web.

The nonwoven web may further comprise further components such as synthetic fibers, natural fibers (e.g. cellulose fibers), or combinations thereof. Synthetic fibers that may be comprised by such nonwoven webs may be polyolefin fibers (e.g. polyethylene, polypropylene, or combinations thereof), or PET fibers, or combinations of polyolefin and PET fibers.

The non-SAF components of the nonwoven web may be removed prior to subjecting the SAF to the method of the present disclosure, however, this does not need to be the case. Instead, the nonwoven web comprising the SAF may be subjected to the method of the present disclosure (as complete nonwoven web, cut, shred or milled) together with the non-SAF components of the nonwoven web.

Prior to or during the method of the present disclosure, the nonwoven web may be cut into pieces or may be milled or otherwise shred to make the SAF more accessible to the oxidated water-soluble salt. Alternatively, the nonwoven web is subjected "as is" to the method.

In one embodiment, the feed stream comprises SAF. In another embodiment, the feed stream comprises SAF and water. In yet another embodiment, the feed stream comprises SAF and ethylene glycol (EG). In even yet another embodiment, the feed stream comprises SAF, water, and ethylene glycol. The water in the feed stream can be RO water, regular tap water, or water containing dissolved inorganic salts at various salt concentrations. A non-limiting example of water with salt is a 0.9 wt % solution of sodium chloride. Other salts with monovalent cations, but higher ionic strength, can be used to reduce the viscosity of the feed stream or alternatively to enable higher SAF concentration to be used. A non-limiting example of a viscosity reducing salt is sodium sulfate.

The feed stream can also comprise any free radical producing chemical compound. Non-limiting examples of such chemical compounds are hydrogen peroxide (H_2O_2), persulfate (such as, sodium persulfate or potassium persulfate), perborate, perphosphate, percarbonate, diazo compounds, ozone, organic free radical initiators (e.g. di-tert-butyl peroxide (DTBP)), combinations thereof, etc. In one embodiment, the feed stream comprises SAF and H_2O_2 . The feed stream can also comprise UV photocatalysts, such as titanium dioxide (TiO_2), etc.

In one embodiment, the feed stream comprises SAF and H_2O_2 . In another embodiment, the feed stream comprises SAF and a H_2O_2 solution.

In one embodiment, the feed stream comprises SAF at a concentration greater than about 1 wt %. In another embodiment, the feed stream comprises SAF at a concentration greater than about 5 wt %. In yet another embodiment, the feed stream comprises SAF at a concentration greater than about 10 wt %. In even yet another embodiment, the feed stream comprises SAF at a concentration of about 2.5 wt %. In one embodiment, the feed stream comprises SAF at a concentration of about 5 wt %.

In one embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 2.5 wt %, the concentration of the H_2O_2 solution is 97.5 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is less than about 3 wt %. In another embodiment, the feed comprises SAF and a H_2O_2 , and the concentration of the SAF is about 5 wt %, the concentration of the H_2O_2 solution is about 95 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is less than about 3 wt %. In yet another embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 2.5 wt %, the concentration of the H_2O_2 solution is 97.5 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 3 wt %. In another embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 5 wt %, the concentration of the H_2O_2 solution is about 95 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 3 wt %.

In one embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 2.5 wt %, the concentration of the H_2O_2 solution is 97.5 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 0.3 wt %. In another embodiment, the feed comprises SAF and a H_2O_2 , and the concentration of the SAF is about 5 wt %, the concentration of the H_2O_2 solution is about 95 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 0.3 wt %. In yet another embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 2.5 wt %, the concentration of the H_2O_2 solution is 97.5 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 0.03 wt %. In another embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the SAF is about 5 wt %, the concentration of the H_2O_2 solution is about 95 wt %, and the concentration of the H_2O_2 in the H_2O_2 solution is about 0.03 wt %.

In one embodiment, the feed comprises SAF and a H_2O_2 solution, and the concentration of the H_2O_2 in the H_2O_2 solution is less than about 3 wt %. In another embodiment, the feed comprises SAF and H_2O_2 , and the concentration of the H_2O_2 in the H_2O_2 solution is less than about 0.3 wt %. In yet another embodiment, the feed comprises SAF and H_2O_2 solution, and the concentration of the H_2O_2 in the H_2O_2 solution is less than about 0.03 wt %.

The viscosity of the feed stream is typically measured with a parallel plate fixture in oscillatory mode, and the complex viscosity reported typically corresponds to a frequency of 1 rad/s. Depending on the SAF concentration the complex viscosity of the feed stream can be higher than 200

Pa·s (or equivalently, 200,000 cP). The feed stream can be in the form of a solution or gel, depending on the concentration of SAF.

The non-renewable energy use (NREU) to make acrylic acid (AA) from the fossil-derived propylene is estimated to be about 50 MJ/kg AA. Therefore, any successful recycling attempt of SAF desirably expends less energy than the NREU to make AA, i.e., less than about 50 MJ/kg SAF for competing environment friendly and commercially with fossil-derived virgin materials. For the purposes of the NREU, it is assumed that the SAF is fully non-neutralized (DN=0).

III. Flow System

Typically, the feed stream is in fluid communication with the flow system via a tube or a channel, and a pump. Non-limiting examples of tubes or channels are glass tubes, metal tubes, alloy tubes (such as, stainless steel tubes), and polymer tubes. The tube or channel can have any cross-sectional shape, such as, circular, rectangular, oval, rhombic, etc. Also, the size of the cross-sectional area of the tube or channel can be the same or vary along the flow direction. A non-limiting example of a varying cross-sectional shape of a tube is an undulating tube that can cause the feed stream to experience extensional stresses as it flows down the tube. These extensional stresses might be beneficial to the degradation of the SAF that is part of the feed stream. Shear stresses can also be beneficial to the degradation of the SAF. Extensional and/or shear stresses can be imparted onto the feed stream by static mixers or other mixing elements placed inside the tube and/or channel that the feed stream flows through.

In one embodiment, the wall shear stress T_w is at least about 0.1 kPa. In another embodiment, the wall shear stress T_w is at least about 1 kPa. In yet another embodiment, the wall shear stress T_w is at least about 5 kPa. In even yet another embodiment, the wall shear stress T_w is at least about 10 kPa.

Non-limiting examples of pumps are centrifugal pumps (such as, axial, radial, and mixed flow pumps) and positive displacement pumps (such as, reciprocating, rotary, piston, diaphragm, gear, peristaltic, screw, and vane). The flow system can employ one or more pumps.

IV. UV Irradiation

The UV irradiation can take place by exposing the flow system to either the sun, or any typical lab or plant UV system, or a combination of both. Typical sources of UV in the lab or plant are LEDs, lasers, gas-discharge lamps, incandescent lamps, etc. The UV radiation degrades SAF into soluble polymers. A typical additional benefit of the UV radiation is its sanitization and sterilization effects on the SAF feed and product streams.

The feed stream gets irradiated by the UV (i.e., gets exposed to the UV radiation) as it flows in the irradiation zone, and it has a residence time in the irradiation zone. In one embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 20 min. In another embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 10 min. In yet another embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 5 min. In one embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 120 s. In another embodiment, the residence time of the feed stream in the UV irradiation zone

is less than about 60 s. In yet another embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 30 s. In even yet another embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 15 s. In one embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 10 s. In another embodiment, the residence time of the feed stream in the UV irradiation zone is less than about 5 s.

The UV total energy is the electric energy that is supplied to the UV unit, and the UV irradiation energy (or simply, UV energy) is the UV energy dissipated in the feed stream. The calculations for the UV total energy and UV irradiation energy are exemplified in the Methods section VIII (as they are well known to those skilled in the art).

In one embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 30 MJ/kg SAF. In another embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 20 MJ/kg SAF. In yet another embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 10 MJ/kg SAF. In even yet another embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 5 MJ/kg SAF. In one embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 1 MJ/kg SAF. In another embodiment, the UV irradiation energy used to convert SAF to soluble polymers is less than about 0.5 MJ/kg SAF.

In one embodiment, the UV total energy used to convert SAF to soluble polymers is less than about 50 MJ/kg SAF. In another embodiment, the UV total energy used to convert SAF to soluble polymers is less than about 32 MJ/kg SAF. In yet another embodiment, the UV total energy used to convert SAF to soluble polymers is less than about 16 MJ/kg SAF. In even yet another embodiment, the UV total energy used to convert SAF to soluble polymers is less than about 10 MJ/kg SAF. In one embodiment, the UV total energy used to convert SAF to soluble polymers is less than about 2 MJ/kg SAF.

The UV irradiation can take place at room temperature or any other temperature. Also, the UV irradiation can take place at atmospheric pressure, in vacuum, or at elevated pressure. Furthermore, the UV irradiation can be preceded or followed by other processes, such as microwave heating, IR heating, ultrasonic irradiation/cavitation, extrusion, extensional stretching, etc.

V. Product Stream

The UV irradiation of the feed stream produces a product stream at the end of the irradiation zone. In one embodiment, the product stream comprises soluble polymers. In another embodiment, the product stream comprises soluble polymers and SAF.

In one embodiment, the soluble polymers have a weight-average molecular weight less than about 5,000,000 g/mol. In another embodiment, the soluble polymers have a weight-average molecular weight less than about 2,000,000 g/mol. In yet another embodiment, the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol. In even yet another embodiment, the soluble polymers have a weight-average molecular weight less than about 500,000 g/mol. In one embodiment, the soluble polymers have a weight-average molecular weight less than about 300,000 g/mol. In another embodiment, the soluble polymers have a weight-average molecular weight less than about 200,000 g/mol. In yet another embodiment, the

11

soluble polymers have a weight-average molecular weight less than about 100,000 g/mol.

In one embodiment, the soluble polymers have a weight-average molecular weight between about 1,000,000 g/mol and about 5,000,000 g/mol. In another embodiment, the soluble polymers have a weight-average molecular weight between about 500,000 g/mol and about 2,000,000 g/mol. In yet another embodiment, the soluble polymers have a weight-average molecular weight between about 100,000 g/mol and about 1,000,000 g/mol. In even yet another embodiment, the soluble polymers have a weight-average molecular weight between about 150,000 g/mol and about 500,000 g/mol. In one embodiment, the soluble polymers have a weight-average molecular weight between about 90,000 g/mol and about 300,000 g/mol. In another embodiment, the soluble polymers have a weight-average molecular weight between about 20,000 g/mol and about 200,000 g/mol. In yet another embodiment, the soluble polymers have a weight-average molecular weight between about 10,000 g/mol and about 100,000 g/mol.

In one embodiment, the soluble polymers have a polydispersity index (PDI) less than about 10. In another embodiment, the soluble polymers have a PDI less than about 6. In yet another embodiment, the soluble polymers have a PDI less than about 4. In even yet another embodiment, the soluble polymers have a PDI less than about 2. PDI is the ratio of the weight-average molecular weight to the number-average molecular weight, and these molecular weights are measured by GPC (described in the Methods section VIII) as it is known to those skilled in the art.

The viscosity of the product stream is typically measured with either a parallel plate fixture in oscillatory mode or a cup and bob fixture in steady mode. The oscillatory viscosity reported typically corresponds to 1 rad/s, and the steady viscosity reported typically corresponds to a shear rate of 4 s⁻¹. Depending on the soluble polymers' concentration and molecular weight, the viscosity of the product stream can be as low as 1 mPa·s (or equivalently, 1 cP; i.e., the viscosity of water).

The ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity reduction ratio (or simply, viscosity ratio). It indicates the extent of the SAF degradation to soluble polymers by the UV flow system. The negative decadic logarithm of the viscosity ratio measures the orders of magnitude change between the viscosity of the feed stream and the product stream. In one embodiment, the feed stream has a viscosity; the product stream has a viscosity; the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and the negative decadic logarithm of the viscosity ratio is less than about 6. In another embodiment, the feed stream has a viscosity; the product stream has a viscosity; the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and the negative decadic logarithm of the viscosity ratio is less than about 4. In yet another embodiment, the feed stream has a viscosity; the product stream has a viscosity; the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and the negative decadic logarithm of the viscosity ratio is less than about 2.

Soluble polymers from the product stream can be derivatized into materials for various applications, such as, adhesives, coatings, water treatment, etc. In one embodiment, soluble polymers from the product stream, either as is or derivatized, are used as an adhesive. In yet another embodiment, soluble polymers from the product stream, either as is or derivatized, are used in fabric care applications. In even

12

yet another embodiment, soluble polymers from the product stream, either as is or derivatized, are used in water treatment applications.

In one embodiment, soluble polymers from the product stream are used as a ply glue in paper products. In another embodiment, soluble polymers from the product stream are used as a ply glue in paper towel products. In yet another embodiment, soluble polymers from the product stream are used as a ply glue in toilet paper products. In even yet another embodiment, soluble polymers from the product stream are used as ply glue in paper products has M_w greater than about 350 kDa. In one embodiment, soluble polymers from the product stream are used as ply glue in paper products has M_w between about 400 kDa and about 500 kDa.

In another embodiment, soluble polymers from the product stream are used as a glue between the paper core and paper towel products. In even another embodiment, soluble polymers from the product stream are used as a glue between the paper core and toilet paper products.

Soluble polymers can be extracted from the product stream via a number of processes. Non-limiting examples of these processes are water evaporation, soluble polymers filtration, water extraction, etc. Also, salts present in the product stream from the use of SAF in AHPs can be removed via any desalination technique known to those skilled in the art. Non-limiting examples of desalination processes are membrane processes (e.g. reverse osmosis, forward osmosis, electrodialysis reversal (EDR), nanofiltration, etc.), freezing desalination, solar desalination, geothermal desalination, ion exchange, wave powered desalination, etc.

VI. Recycled SAF

Soluble polymers from the product stream can be fed into the process to make SAF, thus producing recycled SAF. In one embodiment, the soluble polymers are used to produce a recycled SAF.

In one embodiment, the recycled SAF has a Centrifuge Retention Capacity CRC as measured using the CRC test method as described herein, and wherein the CRC is between about 7 g/g and about 45 g/g. In another embodiment, the recycled SAF has a CRC, and wherein the CRC is between about 10 g/g and about 35 g/g. In yet another embodiment, the recycled SAF has a CRC, and wherein the CRC is between about 15 g/g and about 35 g/g.

VII. Methods

NMR Content Method

The NMR Content Method is used to determine the ratio on a molar basis of functional groups with different NMR signals such as alkene terminal moieties, alkoxy groups (—O—CHR₁— with R₁ one of the group H, alkyl, aryl, heteroaryl, alkoxy or halogene), aliphatic groups (—CHR₁— with R₁ one of the group H, alkyl, aryl, heteroaryl or halogene) and/or other ¹H-NMR active groups.

In this method, proton NMR spectroscopy is used to analyze a sample of soluble material in deuterated water, and peaks of different ¹H-NMR domains are integrated, and ratioed to determine the mole percent of protons of the functional groups or different ¹H-NMR domains, respectively.

The soluble polymer is dried in a vacuum oven (Heraeus Vacutherm type, Thermo Scientific™) at 40° C. and pressure

13

between 10 and 50 mbar for 3 hours. Any small molecular weight alcohol, ester or ether is removed from the soluble polymer.

A flowable solution less than 10% by weight sample dissolved in D₂O and pH adjusted to 5.5-6.5 is prepared. The solution is transferred to an NMR glass grade tube and placed in the sample holder (bore) of a proton NMR instrument. An example of a suitable instrument is a Bruker NMR device with 400 MHz field strength. Instruments of other makes and other field strengths, even including "low-field" instruments operating as low as 60 MHz, can successfully be used to perform this method. A noesy-presat sequence is used to acquire the data and suppress the residual water signal. One of skill will be familiar with appropriate choice of other specific data collection parameters. Appropriate parameters used with the exemplary 400-MHz Bruker instrument above are: acquisition time (FID length) of 4.1 s, relaxation time of 8 s, 90-degree pulse widths, spectral width of 20 ppm, 64 k points in the FID, and 64 repetition scans used. In the Fourier transform step, exponential apodization is used with 0.3-Hz line broadening, and the spectrum is phased into absorption. A spline baseline correction is used to ensure flat baseline on either side of peaks to be integrated.

The following peak domains are typically used for the content determination and integrated:

- 1) One of two terminal alkene protons at chemical shifts in the range of appr. 5-6 ppm, typically one terminal alkene proton at appr. 5.35 ppm $\pm$ 0.5 ppm. (To confirm the identify of such proton peak as terminal alkene proton a standard edited ¹H-¹³C HSQC sequence can be used (following e.g. W. Willker, D. Leibfritz, R. Kerssebaum & W. Bermel, Magn. Reson. Chem. 31, 287-292 (1993)) to determine that the alkene signals seen in the 1D-1H spectrum are both attached to the same methylene (secondary) carbon ($\text{—CH}_2\text{—}$). The resulting integral is called "integral_alkene".
- 2) alkoxy protons at chemical shifts in the range of appr. 3.2-3.8 ppm, typically at appr. 3.6 ppm. The resulting integral is called "integral_3.6". In case more than one signal appears at appr. 3.6 ppm, i.e. in the range of appr. 3.2-3.8 ppm, the signal with the largest integral is chosen to get "integral_3.6".
- 3) methylene protons of aliphatic CH-groups, typically at ca. 1.5 ppm and 2.1 ppm, or at ca. 1.8 ppm. The resulting integral is called "integral CH".
- 4) other aliphatic groups at chemical shifts in the range of appr. 1.0-2.6 ppm.
- 5) other groups or peaks of different ¹H-NMR domains can be analyzed as well, provided that they
 - a) are separated signals,
 - b) have peak maxima are at a distance of at least 0.5 ppm.
- 6) The peaks in the NMR spectrum corresponding to class 1), 2), 3) and/or 4) are identified and, if present, integrated. If no such peak is present, this reported as no measurable content of class 1), 2), 3) or 4), respectively. As known by the person skilled in the art, the integration ranges from baseline (start of signal) to baseline (end of signal), or in case of broad and/or complex the boarders of integration result from the start of the next neighbouring signal.

The ratio "ratio 3.6:CH" is calculated via the following formula:

$$\text{ratio 3.6:CH} = \text{integral_3.6} / \text{integral_CH}$$

14

ratio 3.6:CH is a unitless number and represents the ratio of protons at ca. 3.6 ppm

The content of alkene "% alkene" is calculated via the following formula:

$$\% \text{ alkene} = [\text{integral_alkene} / (\text{integral_alkene} + \text{integral_3.6} + \text{integral_CH})] * 100\%$$

The content of methylene "% CH" is calculated via the following formula:

$$\% \text{ CH} = [\text{integral_CH} / (\text{integral_alkene} + \text{integral_3.6} + \text{integral_CH})] * 100\%$$

The content of protons at about 3.6 ppm (such as alkoxy protons) "%3.6 ppm" is calculated via the following formula:

$$\%3.6 = [\text{integral_3.6} / (\text{integral_alkene} + \text{integral_3.6} + \text{integral_CH})] * 100\%$$

The ratio "ratio 3.6:CH" is reported to the nearest 0.001.

The content of alkene, methylene and protons at about 3.6 ppm are reported in % to the nearest 0.01%.

Centrifuge Retention Capacity (CRC) Test Method

Capacity of the superabsorbent fibers is determined according to the Centrifuge Retention Capacity (CRC) test method as set out in EDANA NWSP 241.0.R2(19). In deviation from EDANA NWSP 241.0.R2(19) the sampling (chapter 8 in EDANA NWSP 241.0.R2(19)) for the superabsorbent fibers, superabsorbent nonwovens and/or the superabsorbent core is as following: The superabsorbent fibers, superabsorbent nonwovens and/or the superabsorbent core are cut into pieces with approximately 5 mm as largest dimension. The cutting can e.g. be done manually with scissors. Care is taken that the fibrous structure (the core, the nonwoven or the bulk of fibers) is not majorly compressed before or during the cutting process. This ensures sufficient void space between the superabsorbent fibers, so they can be predominately wetted by the swelling medium at the entire surface area.

Further deviations from or additions to EDANA NWSP 241.0.R2(19) in the procedure (chapter 9.1-9.5 in EDANA NWSP 241.0.R2(19)) for the superabsorbent fibers, superabsorbent nonwovens and/or the superabsorbent core are as following:

The sample for the measurement is taken carefully, e.g. with a lab pincet, to put it into the teabag. With a lab pincet, the fibers are carefully distributed in the teabag to avoid lumps and fiber lumps, if any, are carefully opened.

When sealing the teabag, care is taken that no material of the superabsorbent fibers, superabsorbent nonwovens and/or the superabsorbent core is in the area of the seal. This ensures a complete and sufficiently strong sealing of the teabag.

All other items of the test method are executed as set out in EDANA NWSP 241.0.R2(19).

UV Total Energy Calculations

The UV total energy is the electric energy that is supplied to the UV unit, and is based on the voltage and amperage of the UV unit, and the residence time of the feed stream. If the voltage of the UV unit is 200 V and the amperage is 16 A, then the power of the UV unit is 3200 W. The UV total energy per unit mass of SAF is calculated based on the UV total power and the amount of SAF that is in the feed stream covering the whole length of the UV lamp. If the flow system tube has a diameter of 0.368 cm then, the number of flow system tubes that fit in the 15 cm length of the UV lamp is 40. If the flowrate of the feed stream in each tube is 5 mL/min and the SAF concentration is 2.5 wt %, then the UV total energy per unit mass of SAF is calculated as 3200

15

$(\text{J/s}) \times 60 (\text{s}) / (40 \times 5 (\text{mL feed}) \times 0.025 (\text{g SAF/g feed}) \times 1 (\text{g feed/mL feed})) = 38.4 \text{ MJ/kg SAF}$, assuming the density of the feed stream is 1 g/mL.

UV Irradiation Energy Calculations

The UV irradiation energy (or simply, UV energy) is the UV energy dissipated in the feed stream, is based on the total energy density of the lamp and is calculated as shown in the example below (well known to those skilled in the art). The total energy density was measured with the UV Power-MAP® #20082105 A/B/C/V (EIT, Inc., Sterling, VA) running under the UV lamp at various speeds. For example, at 5.6 m/min speed the energy densities were: UVA 488 mJ/cm², UVB 466 mJ/cm², and UVV 102 mJ/cm², yielding a total energy density of 1,056 mJ/cm². The energy density of UVV is neglected in these calculations. Similarly, at 11.2 m/min speed, the respective energy densities were: 249, 238, and 52 mJ/cm², and the total energy density was 539 mJ/cm². Extrapolation is used for speeds outside the above range of speeds, and interpolation is used for speeds inside the above range of speeds.

In the example used in the UV Total Energy Calculations above, the flowrate of the feed stream through the 0.368 cm diameter tube was 5 mL/min. Therefore, the average linear speed of the feed stream is calculated as $5 (\text{cm}^3/\text{min}) / ((\pi \times (0.368 \text{ cm})^2 / 4) \times 100) = 0.47 \text{ m/min}$. This speed is outside the range measured above, and thus we did an extrapolation to calculate the total energy density at 0.47 m/min as 12,565 mJ/cm². The UV irradiation energy per unit mass of SAF is then calculated as: $12,565 (\text{J/s} \cdot \text{cm}^2) \times 0.368 (\text{cm}) \times 15 (\text{cm}) / (0.025 (\text{g SAF/g feed}) \times \pi \times ((0.368^2 (\text{cm}^2)) / 4) \times 15 \text{ cm} \times 1 \text{ g feed/cm}^3) = 1.7 \text{ MJ/kg SAF}$.

Note that in the examples above the UV total energy is 38.4 MJ/kg SAF and the corresponding UV irradiation energy is 1.7 MJ/kg SAF. These UV energy values indicate that the energy efficiency of the UV unit is $1.7/38.4 = 4.4\%$.

Molecular Weight Distribution (MWD) Analysis

It is done using the Tetradection Gel Permeation Chromatography (GPC-T; Jordi Labs; Mansfield, MA) technique, as it is well known to those skilled in the art. The technique uses a GPC column, refractive index (RI) detector, right angle light scattering (RALS) detector, and viscometry (DP) detector. Approximately 50 mg of a sample are diluted into 2 mL of 0.1M NaNO₃ with 0.01M Na₂HPO₄ with gentle agitation yielding transparent solutions. These solutions are injected without further processing. The instrument is calibrated with a Pullulan standard ($M_w = 61,330 \text{ Da}$), and its performance is verified with a Dextran standard ($M_w = 68,162 \text{ Da}$).

The foregoing description is given for clearness of understanding only, and no unnecessary limitations should be understood therefrom, as modifications within the scope of the invention may be apparent to those having ordinary skill in the art.

The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as "40 mm" is intended to mean "about 40 mm."

Every document cited herein, comprising any cross referenced or related patent or application, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references,

16

teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. A method for degrading superabsorbent fibers (SAF) to soluble polymers, wherein the soluble polymers comprise from 5 weight-% to 75 weight-% of polymerized acrylic acid monomer units based on the total weight of the soluble polymers, the method comprising:

flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV, and producing a product stream comprising the soluble polymers at the end of the irradiation zone; wherein the feed stream comprises the SAF at a concentration greater than about 1 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; and wherein the UV used to convert the SAF to the soluble polymers requires a UV total energy of less than about 50 MJ/kg SAF.

2. The method of claim 1, wherein the residence time is less than about 60 s.

3. The method of claim 1, wherein the UV total energy is less than about 16 MJ/kg SAF.

4. The method of claim 1, wherein the feed stream comprises the SAF and water.

5. The method of claim 1, wherein the feed stream comprises the SAF and hydrogen peroxide.

6. The method of claim 1, wherein the SAF has a sodium level as amount of Na of more than 8 wt %.

7. The method of claim 1, wherein the SAF has a sodium level as amount of Na of between about 10 wt % and about 20 wt %.

8. The method of claim 1, wherein the feed stream has a viscosity; wherein the product stream has a viscosity; wherein the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and wherein the negative decadic logarithm of the viscosity ratio is less than about 6.

9. The method of claim 1, wherein the feed stream has a viscosity; wherein the product stream has a viscosity; wherein the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and wherein the negative decadic logarithm of the viscosity ratio is less than about 4.

10. The method of claim 1, wherein the feed stream has a viscosity; wherein the product stream has a viscosity; wherein the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and wherein the negative decadic logarithm of the viscosity ratio is less than about 2.

11. The method of claim 1, wherein the soluble polymers have a weight-average molecular weight less than about 2,000,000 g/mol.

12. The method of claim 1, wherein the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol.

17

13. The method of claim 1, wherein the soluble polymers have a polydispersity index (PDI) less than about 4.

14. The method of claim 1, wherein the soluble polymers are used to produce a recycled SAF and/or recycled superabsorbent polymer particles (SAP); the recycled SAF and/or SAP comprises soluble polymers at a concentration; and wherein the soluble polymer concentration is less than about 30%.

15. The method of claim 1, wherein the SAF have a Centrifuge Retention Capacity as measured using a CRC test method of at least 15 g/g.

16. A method for degrading superabsorbent fibers (SAF) to soluble polymers, wherein the soluble polymers comprise from 5 weight-% to 75 weight-% of polymerized acrylic acid monomer units based on the total weight of the soluble polymers, the method comprising:

flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV, and producing a product stream comprising the soluble polymers at the end of the UV irradiation zone; wherein the feed stream comprises water and the SAF at a concentration greater than about 1 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; wherein the UV used to convert the SAF to the soluble polymers requires a UV total energy of less than about 16 MJ/kg SAF; and wherein the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol.

17. A method for degrading superabsorbent polymer (SAF) to soluble polymers, wherein the soluble polymers comprise from 5 weight-% to 75 weight-% of polymerized acrylic acid monomer units based on the total weight of the soluble polymers, the method comprising:

flowing a feed stream comprising the SAF into a UV irradiation zone, irradiating the feed stream with UV,

18

and producing a product stream comprising the soluble polymers at the end of the UV irradiation zone; wherein the feed stream comprises water and the SAF at a concentration greater than about 5 wt %; wherein the feed stream has a residence time in the UV irradiation zone of less than about 120 s; wherein the UV used to convert the SAF to the soluble polymers requires a UV total energy of less than about 16 MJ/kg SAF; and wherein the soluble polymers have a weight-average molecular weight less than about 1,000,000 g/mol.

18. The method of claim 17, wherein the feed stream has a viscosity; wherein the product stream has a viscosity; wherein the ratio of the viscosity of the product stream to the viscosity of the feed stream is the viscosity ratio; and wherein the negative decadic logarithm of the viscosity ratio is less than about 4.

19. The method of claim 17, wherein the SAF has a sodium level as amount of Na of between about 10 wt % and about 20 wt %.

20. The method of claim 17, wherein the feed stream further comprises hydrogen peroxide.

21. The method of claim 17, wherein the SAF are provided as loose fibers.

22. The method of claim 17, wherein the SAF are provided in the form of a nonwoven web, wherein the nonwoven web comprises SAF.

23. The method of claim 22, wherein the nonwoven web is cut, shred or milled prior to being provided to the method or while carrying out the method.

24. The method of claim 17, wherein the method is carried out as a continuous process.

25. The method of claim 17, wherein the method is carried out as a batch process.

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